

OMAR KHATAB

Frontend Developer (React / Next.js / TypeScript) | Engineering Software Developer

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[Portfolio](#) | [GitHub](#) | [LinkedIn](#)

SUMMARY

Mechanical Power Engineer turned Frontend Developer with expertise in React, Next.js, TypeScript, and Tailwind CSS. Strong foundation in **Computer Science** including Data Structures & Algorithms, Database (MySQL), and Python. Specialized in building domain-specific MEP engineering tools that automate complex HVAC calculations. Available immediately for Junior Frontend Developer roles in Cairo.

TECHNICAL SKILLS

Frontend Architecture: React.js, Next.js 15 (App Router), TypeScript (Strict), JavaScript ES6+, HTML5, CSS3, Semantic HTML5

UI & Styling: Tailwind CSS, Lucide React, Responsive Design, Flexbox, CSS Grid, BEM

Computer Science: Data Structures & Algorithms, Database (MySQL), Python

Engineering Domain: HVAC Calculations, Duct Sizing (Equal Friction & Velocity), Cooling Load ($Q=U \cdot A \cdot \Delta T$), Airflow (CFM), Thermodynamics, Fluid Mechanics

State & Logic: React Hooks, Context API, Input Validation, Real-time Calculation Engine, Constraint Solving (Aspect Ratio)

APIs & Tools: REST APIs, Git, GitHub, Vercel CI/CD, Chrome DevTools

FEATURED PROJECTS

HVAC Engineering Suite & Interactive Calculators - Featured Project

[Live Demo](#) | [Code](#) | Sep 2026 - Present

- Built high-performance HVAC web app for MEP consultants using Next.js 15 App Router, TypeScript Strict, Tailwind CSS, Lucide React, deployed on Vercel (29 commits, CI/CD).
- Advanced Duct Sizing Suite: Algorithm Selection for duct sizing, computes dimensions from airflow (CFM) and velocity (FPM) using continuity equation $\text{Area} = \text{CFM} / \text{Velocity}$ as base.
- Constraint Controls & Aspect Ratio: Supports custom W:H restrictions to fit ducts within architectural false ceiling limits, validates rectangular dimensions against max height/width, automatically adjusts to maintain required airflow area.
- Space Cooling Load Calculator: Evaluates envelope heat gain via conduction & solar radiation using U-values and ΔT , computes internal gains from occupancy, lighting density (W/m^2), appliances, applies safety factors, outputs BTU/h, kW, TR.

E-Commerce Store

[Live Demo](#) | [Code](#) | Jul - Aug 2026

- Built complete e-commerce storefront using React, React Router, Context API with DummyJSON REST API, loading states, error handling.
- Implemented shopping cart with add/remove/quantity, wishlist, search & category filtering, Framer Motion animations, React Hot Toast feedback.

SaaS Landing Page

[Live Demo](#) | [Code](#) | Nov - Dec 2024

- Developed modern SaaS landing with Next.js App Router, hero, features, pricing, testimonials, FAQ, responsive, smooth scroll animations.

Admin Dashboard Panel - 8 Pages

[Live Demo](#) | [Code](#) | Sep - Oct 2023

- Designed 8-page admin panel (Dashboard, Projects, Courses, Files, Friends, Plans, Profile, Settings) with pure HTML5, CSS3, JavaScript, Flexbox, Grid.

Roots Before React - 5 Templates

[Live Demo](#) | [Code](#) | Jan - Jun 2023

- Built 5 landing pages (Leon, Kasper, Digital, Gonex, Special Design) with pure HTML5/CSS3/JS to master fundamentals before frameworks.

ENGINEERING BACKGROUND

B.Sc. in Mechanical Power Engineering - Ain Shams University (2019 - 2024)

- Graduation Project: Solar Cell Cooling System - Active thermal management achieving reported 15% efficiency improvement.
- Academic Project: Wind Turbine Pitch Control System - Dynamic pitch-control mechanisms.
- Internships: National Authority for Tunnels - HVAC Systems & Ventilation Design; Arab Contractors - IC Engines.

EDUCATION & ADDITIONAL INFORMATION

Degree: B.Sc. Mechanical Power Engineering | **Military Service:** Completed | **Graduation:** 2024

University: Ain Shams University | **Arabic:** Native | **English:** Good